

SAMUEL UMOREN

Machine Learning Engineer / Model Observability & LLM Systems
samuelumoren365@gmail.com • umoren.xyz

Machine Learning Engineer with seven years across software engineering, data systems, and applied AI. Has led the design and delivery of Python-based classification, retrieval, and evaluation systems, with recent ownership spanning model observability, feedback pipelines, evidence-grounded LLM workflows, and asynchronous ML infrastructure.

MACHINE LEARNING & DATA EXPERIENCE

Sift

AI Engineer

Jun. 2025 – Jun. 2026

- Led the design and implementation of Sift's AI evaluation and observability layer, capturing queries, user feedback, retrieved context, classification confidence, embedding activity, and end-to-end latency so the team could diagnose response quality and model behaviour.
- Architected an organisation-specific semantic classifier using sentence-transformer embeddings, cosine similarity, confidence thresholds, and deterministic fallbacks, removing repeated LLM calls from the common path while retaining model fallback for uncertain cases.
- Owned the reliability and access-control layer for Sift's hybrid BM25/vector retrieval, connecting AI answers to exact Slack and GitHub evidence through decision traces, sufficiency gates, private-channel controls, and safeguards against unsupported answers.
- Architected and delivered batch, incremental, and event-driven ingestion for Slack, Notion, Linear, and GitHub using OAuth, webhooks, reconciliation jobs, MongoDB/Neo4j persistence, and asynchronous AWS Batch workflows managed with Terraform.

Klysera

Research & Development Lead (Contract)

Sep. 2025 – May 2026

- Defined Klysera's core intellectual property—the “Ideal Klysera Engineer”—and led four research programs that translated it into competency models, curriculum priorities, product positioning, and engineering career frameworks.
- Architected the research and analytics system behind that thesis: reproducible pandas pipelines comparing 200 current job descriptions from 36 companies with a 50-role historical baseline, plus a self-service Next.js/Supabase dashboard for skill, role-coverage, and curriculum decisions.

ACADEMIC MACHINE LEARNING PROJECT

SMS Spam Classification

Python, NLTK, scikit-learn

- Designed and evaluated SMS spam classifiers using bag-of-words and TF-IDF features across Naive Bayes, logistic regression, and linear, polynomial, and RBF support vector machines; selected the bag-of-words linear SVM with 98.4% accuracy and 0.936 F1 on the held-out test set.

ADDITIONAL EXPERIENCE

Cortex MIPs Working Group

Protocol Researcher

Monad MIP Research Lab

Jun. 2026 – Present

- Established a reusable research lab across Monad Improvement Proposals, designing explicit hypotheses, automated test harnesses, Testnet evidence matrices, and CI regressions; isolated a Foundry compatibility boundary and converted it into a reusable regression fixture.

Cilia & Independent

Technical Documentation Engineer

Jan. 2020 – Dec. 2025

- Independently rebuilt [Mono Connect.js documentation](#) from a long README into a searchable developer site with API references, React, Angular, and Next.js integration guides, and advanced reauthorisation and direct-debit flows.
- Published [more than 200 technical articles](#) for software-engineering audiences.

Fress, Mezovest & JiggleNg

Frontend Engineer

2019 – 2022

- Owned frontend delivery across creator, fintech, and mobile-web products, including a course-platform MVP delivered in two weeks and a Mezovest payment feature that protected 70% of customers from churn during a fintech policy change.

SKILLS

Machine learning & LLM systems: Python, pandas, NumPy, scikit-learn, NLTK, sentence-transformers, LlamaIndex, embeddings, semantic classification, NLP, model evaluation, confidence scoring

Data & retrieval: MongoDB, PostgreSQL, Neo4j, Amazon S3, Supabase, BM25/vector retrieval, batch processing, event-driven pipelines

Infrastructure: AWS Batch, AWS Lambda, Docker, Terraform, GitHub Actions, CI/CD

Software engineering: TypeScript, JavaScript, Node.js, Next.js, React, REST APIs, GraphQL

EDUCATION

University of Uyo

2015 – 2021

Bachelor of Engineering in Computer Engineering